Research Portfolio

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Graduate Research Lead | BCA Graduate (CGPA: 8.79/10)

Focus: Hybrid Research in Cognitive Science & Artificial Intelligence

Technical Milestones

Studylytics

Predictive Cognitive Profiling: Identified a -0.02 correlation between latency and accuracy. Model achieved **95% Precision**.

Click to view Research Report →

Mobile AI Optimization

WRC Internship: Optimized MobileNetV2 to a **3.7MB footprint** with **~30ms latency** via TFLite quantization.

Click to view Technical Summary →

Professional Vision

Operating at the intersection of **AI, Cognitive Science, and Pedagogy**, I develop systems that quantify human cognitive load to build adaptive learning environments. My work aims to bridge the "supervision gap" in digital-native learners by translating behavioral interaction data into actionable pedagogical insights. I am dedicated to building resource-efficient, human-centered technology that enhances how the digital generation learns and adapts.

LinkedIn **GitHub** **Medium**

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Studylytics: Cognitive Analytics

Pilot Study | n=135

This study investigates the intersection of Cognitive Science and AI by modeling human learning behavior using interaction data. The system quantifies cognitive load and performance variability in digital-native learners.

ML Precision
95%

Latency Correlation
-0.02

Reasoning Accuracy
69%

Evaluation
99/100

Key Findings

- **Cognitive Bottleneck:** Reasoning tasks show lowest accuracy (~69%), indicating highest cognitive load.
- **Speed–Accuracy Tradeoff:** Weak negative correlation (-0.02) suggests longer time reflects confusion, not better thinking.
- **Focus Threshold:** Performance peaks at moderate focus (Level 7) and declines under overload.
- **Learning Trend:** Accuracy improves over time (86% → 100%), indicating measurable learning.

Limitations

This is a single-user pilot study. Results may not generalize. A manual stimulus rotation strategy was used to reduce familiarity bias. Future work requires multi-user data and balanced datasets.

Visual Evidence

Fig 1: Accuracy by Cognitive Type

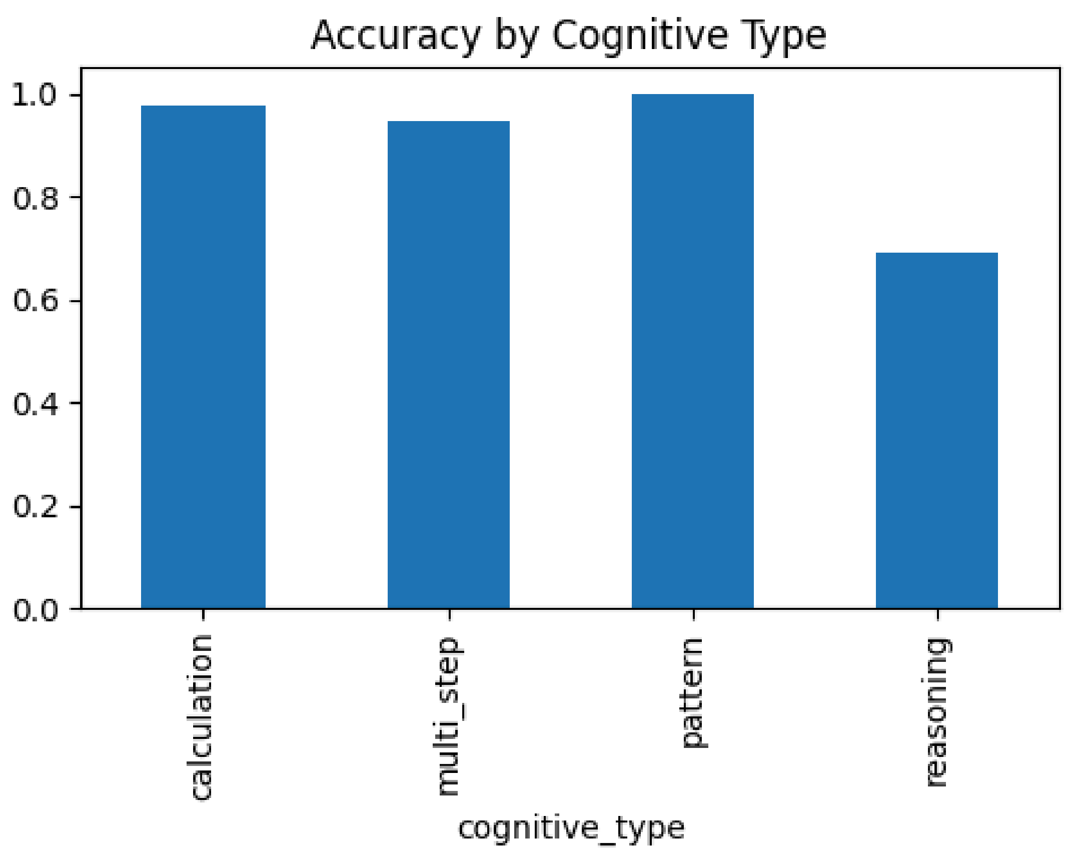
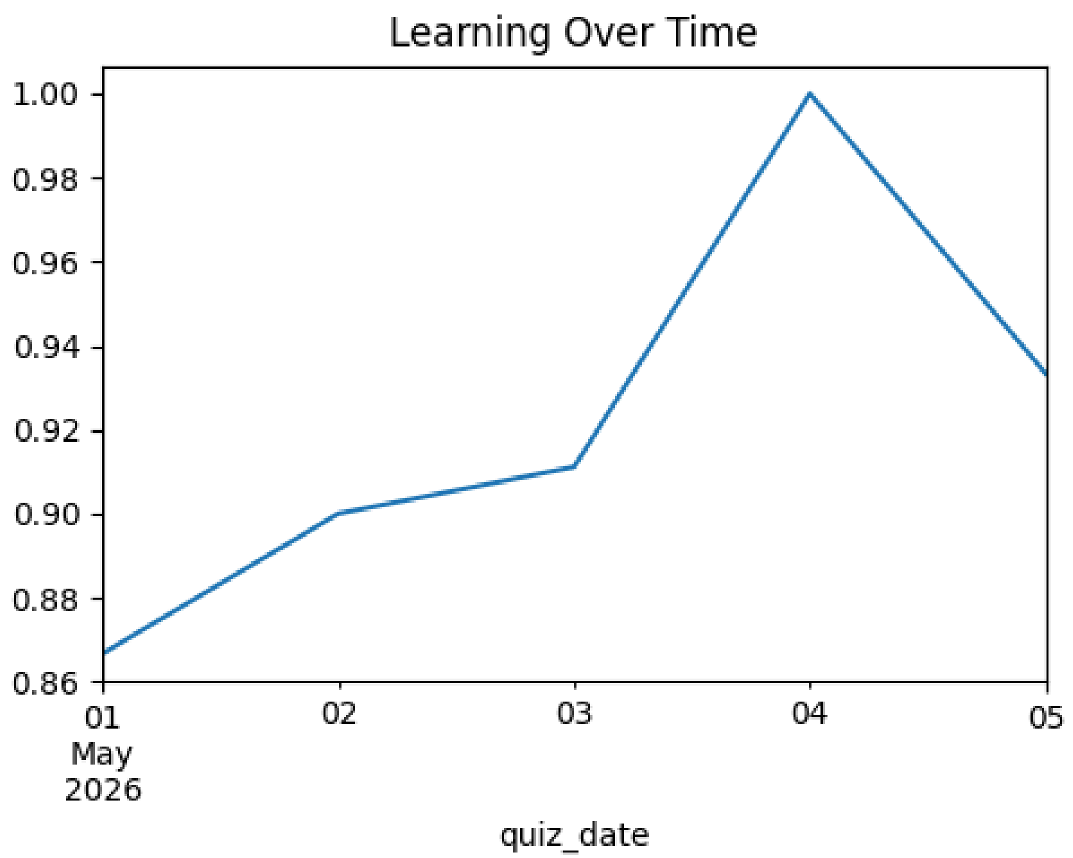


Fig 2: Learning Over Time



Methodology

Data was collected using a Flask–MySQL backend. Analysis was performed using Python (Pandas, Scikit-learn). A Random Forest model was used for prediction, identifying time spent and cognitive type as key features.

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Image Classification Model

Win Research Centre (WRC) | Internship

Architected a production-ready image classification system for the **Fit Choice World (FCW)** app. This project focused on balancing high-precision deep learning with the strict resource constraints of mobile hardware.

MODEL FOOTPRINT 3.7 MB	INFERENCE LATENCY ~30 ms
TEST ACCURACY 97.0%	EVALUATION 100/100

Optimization Engineering

- **Architecture:** Utilized **MobileNetV2** with transfer learning for optimal speed-to-accuracy ratio on mobile edge devices.
- **TFLite Quantization:** Performed post-training quantization to compress the model from standard .h5 to a 3.7MB .tflite format, reducing memory usage while maintaining precision.
- **Data Augmentation:** Successfully resolved initial 6% overfitting in 'Random' and 'Food' categories using rotation, zoom, and horizontal flip techniques.

Visual Evidence

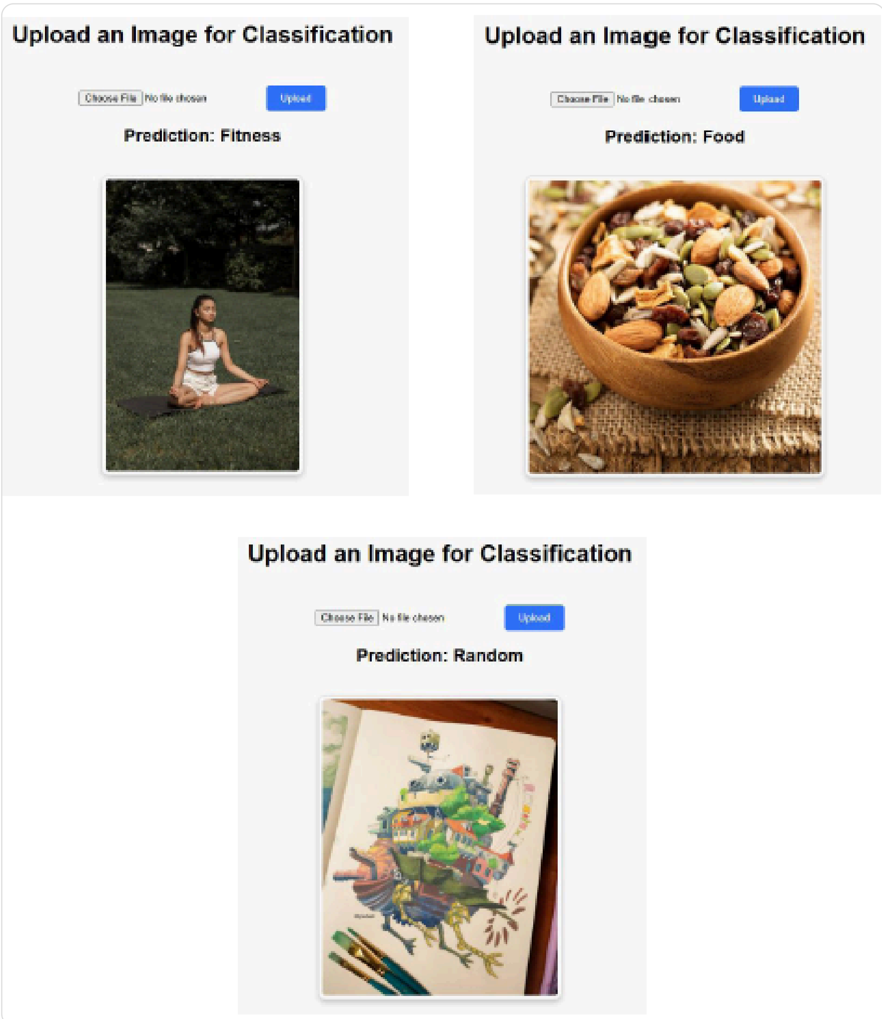

Fig 1: Real-time Flask UI identification showing category prediction and confidence scores.

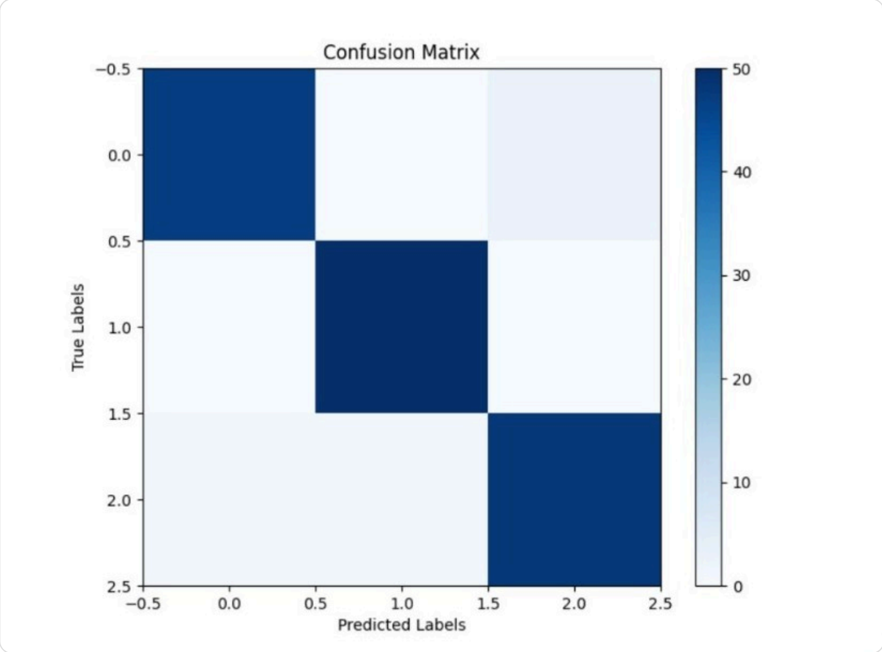

Fig 2: Confusion Matrix validating 97% accuracy and high precision across Food, Fitness, and Random classes.

Technical Implementation

Developed a **Python Flask API** to handle image buffers and return instantaneous predictions. The pipeline was designed to handle high-frequency requests, identifying Food, Fitness, and Random stimuli to automate user health logging.